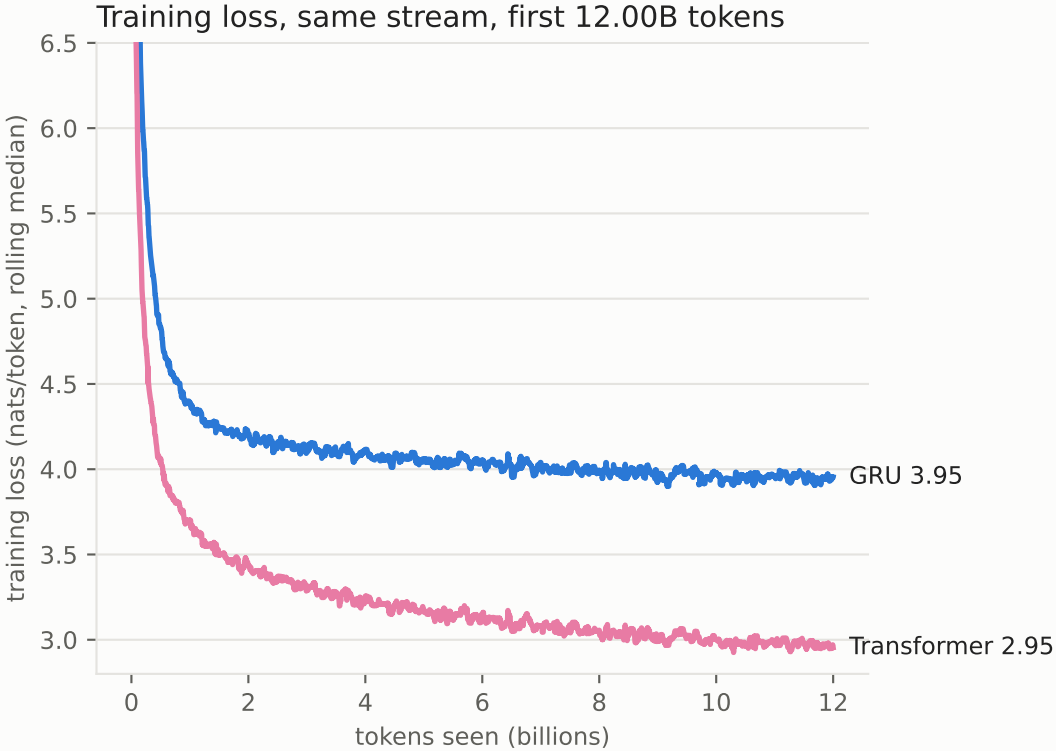
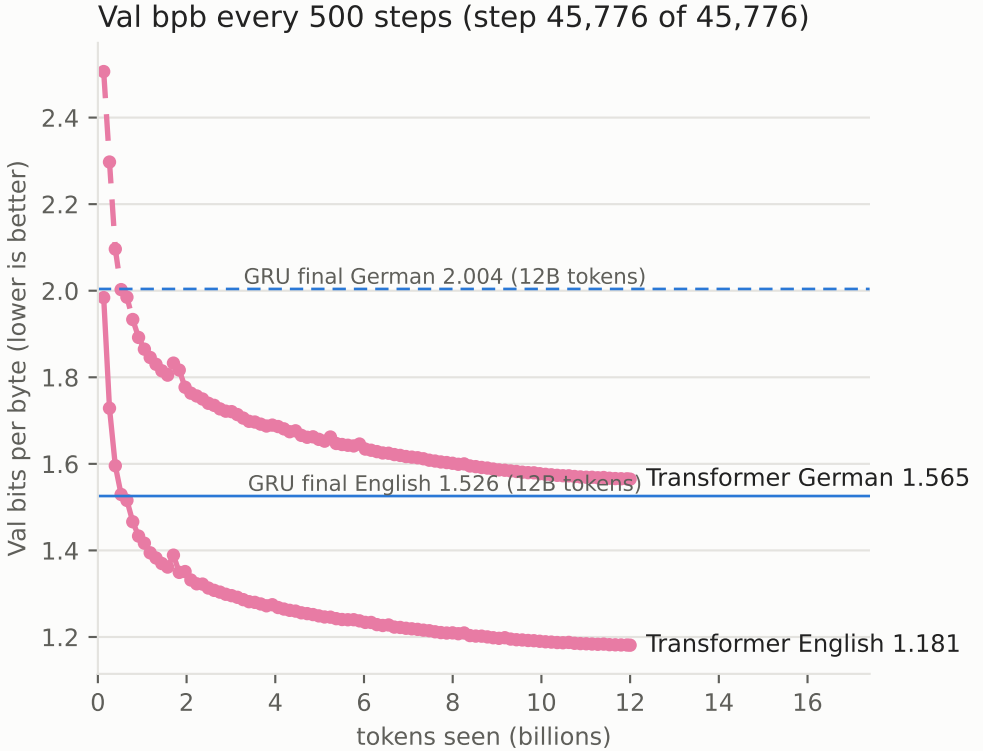

Training loss, same stream, first 12.00B tokens
training loss (nats/token, rolling median)
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
0
2
4
6
8
10
12
tokens seen (billions)
GRU 3.95
Transformer 2.95


Val bpb every 500 steps (step 45,776 of 45,776)
Val bits per byte (lower is better)
2.4
2.2
2.0
1.8
1.6
1.4
1.2
0
2
4
6
8
10
12
14
16
tokens seen (billions)
GRU final German 2.004 (12B tokens)
GRU final English 1.526 (12B tokens)
Transformer German 1.565
Transformer English 1.181